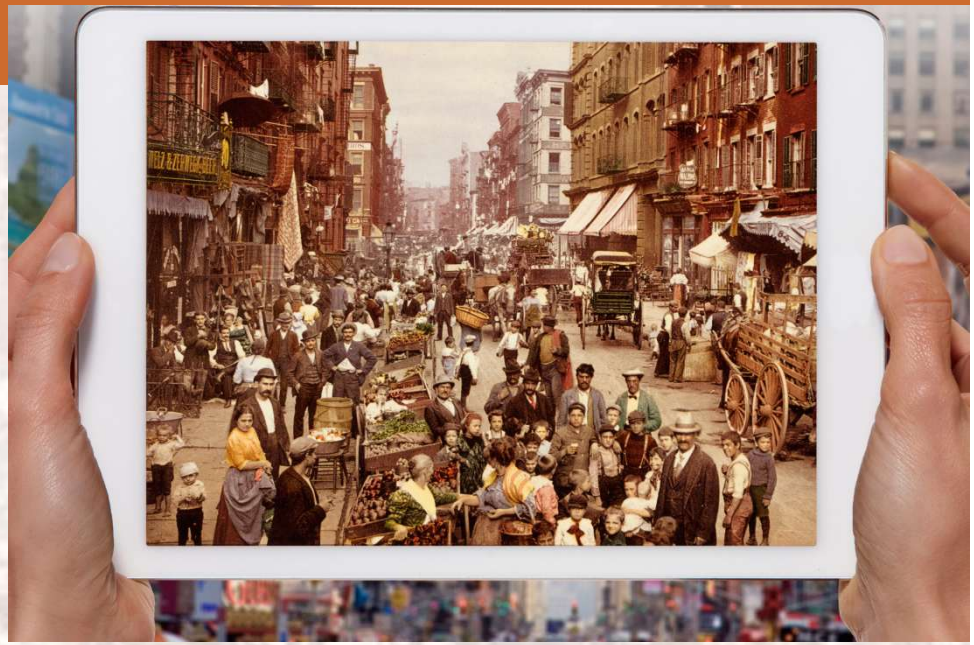
經濟學



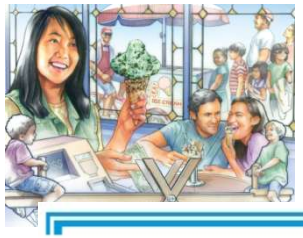
CHAPTER 6

Supply, Demand, and Government Policies

National Central University

Look for the answers to these questions:

- What are price ceilings and price floors? What are some examples of each?
- How do price ceilings and price floors affect market outcomes?
- How do taxes affect market outcomes? How do the effects depend on whether the tax is imposed on buyers or sellers?
- What is the incidence of a tax (租稅歸宿)? What determines the incidence?



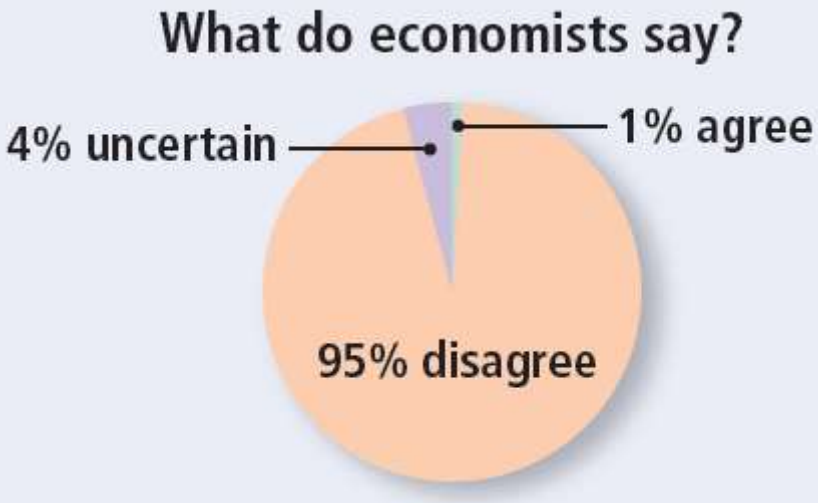
Government Policies That Alter the Private Market Outcome

- Price controls
  - Price ceiling: legal maximum on the price at which a good can be sold
    - Rent-control laws
  - Price floor: legal minimum on the price at which a good can be sold
    - Minimum wage laws
- Taxes: government can make buyers or sellers pay a specific amount on each unit

ASK THE EXPERTS

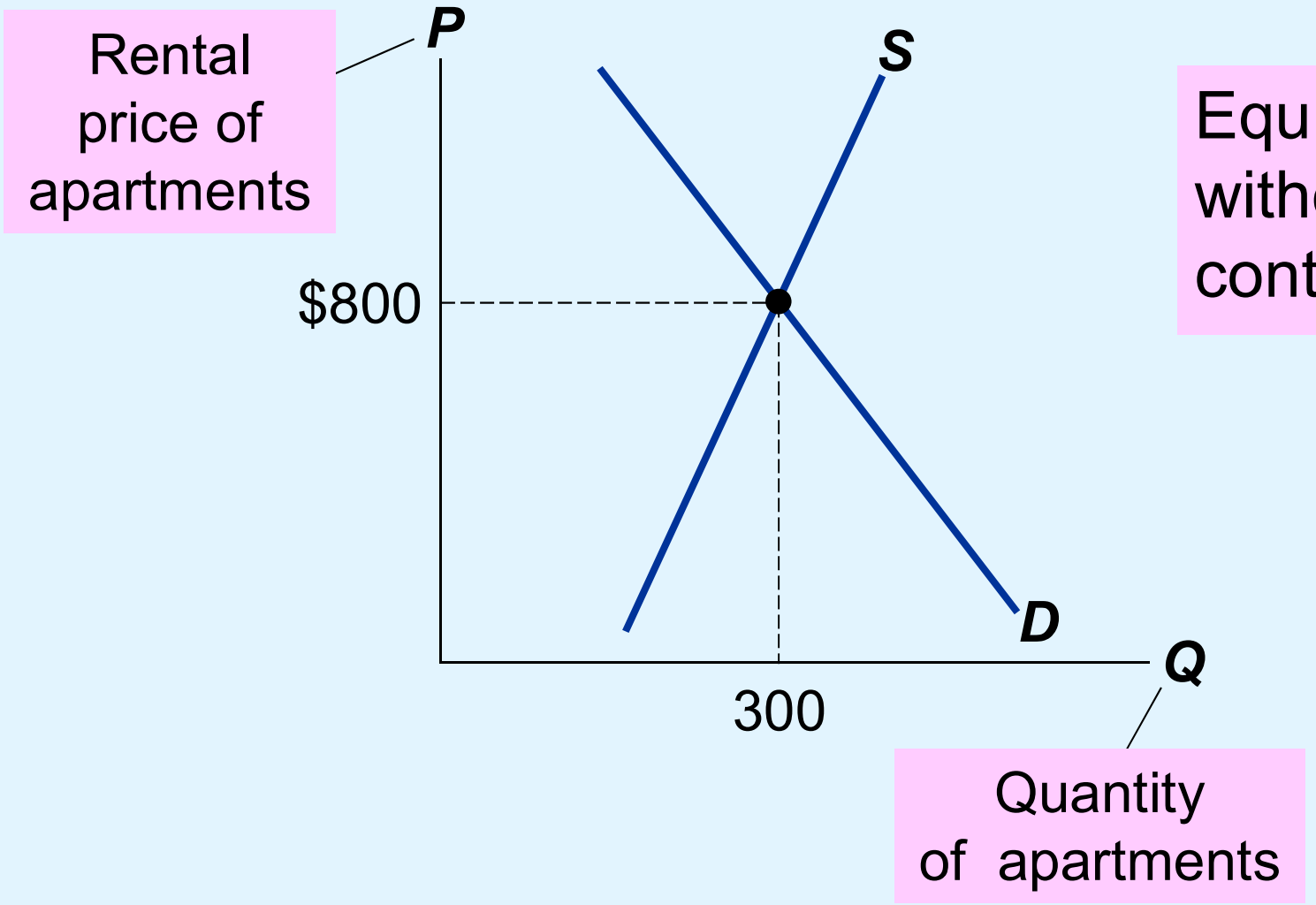
Rent Control

"Local ordinances that limit rent increases for some rental housing units, such as in New York and San Francisco, have had a positive impact over the past three decades on the amount and quality of broadly affordable rental housing in cities that have used them."



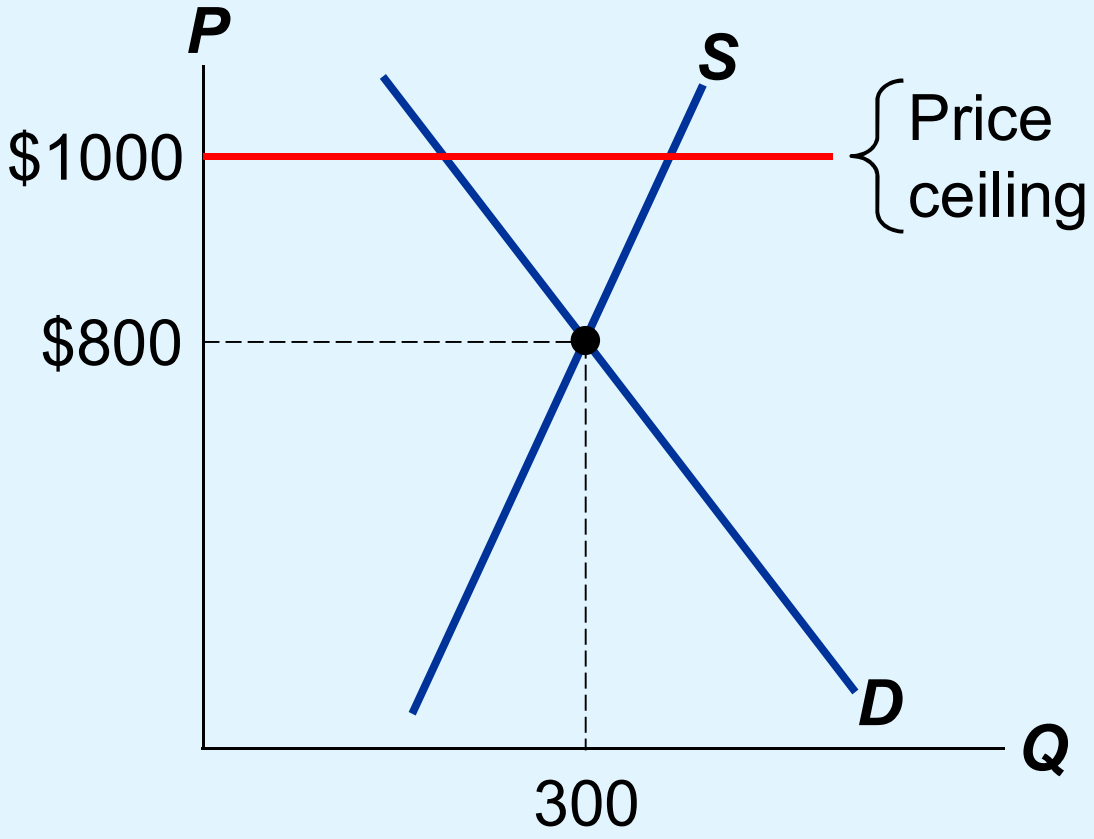
In the US, rent caps limit how much your landlord can INCREASE your rent.

EXAMPLE 1: The Market for Apartments



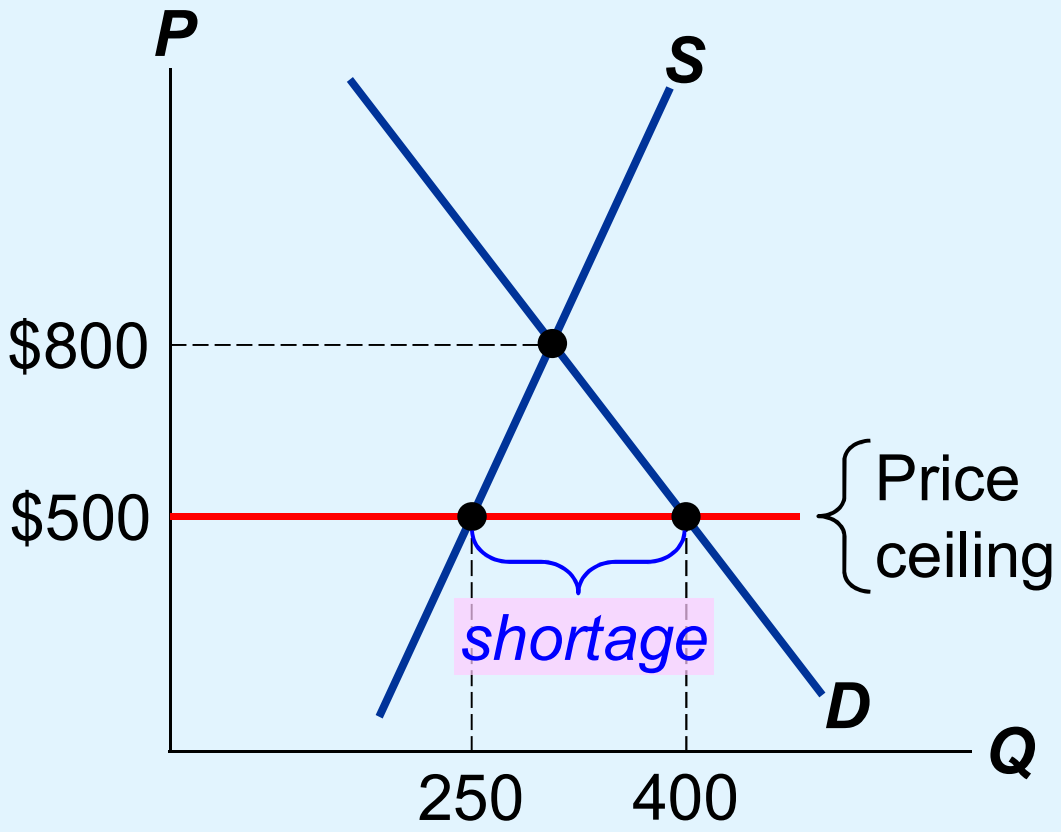
How Price Ceilings Affect Market Outcomes

A price ceiling above the equilibrium price is **not binding**— has no effect on the market outcome.



How Price Ceilings Affect Market Outcomes

The equilibrium price ($800) is above the ceiling and therefore illegal.
The price ceiling is **binding**, causes a shortage.

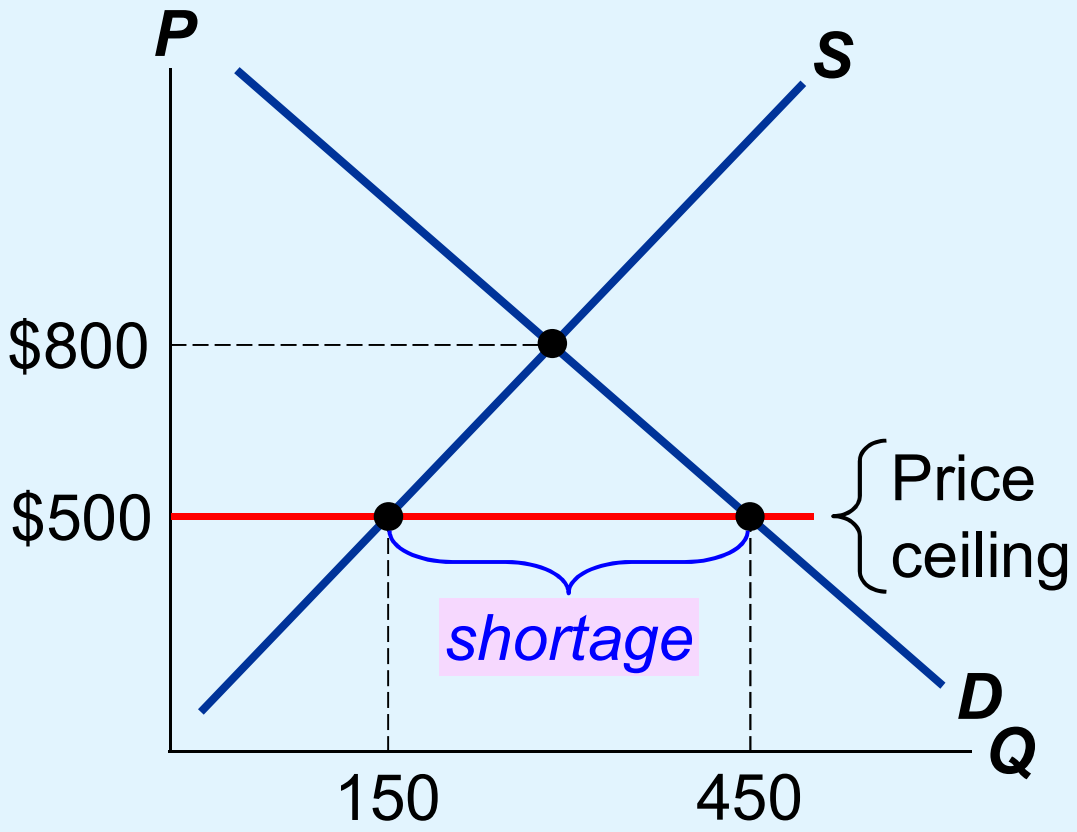


The effects of rent control in the short run (when the supply and demand for housing are inelastic) are: Small shortage and Reduced rents

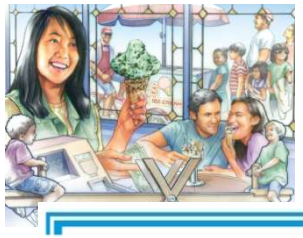
How Price Ceilings Affect Market Outcomes

In the long run, supply and demand of rental apartments are more price-elastic.

So, the shortage is larger.



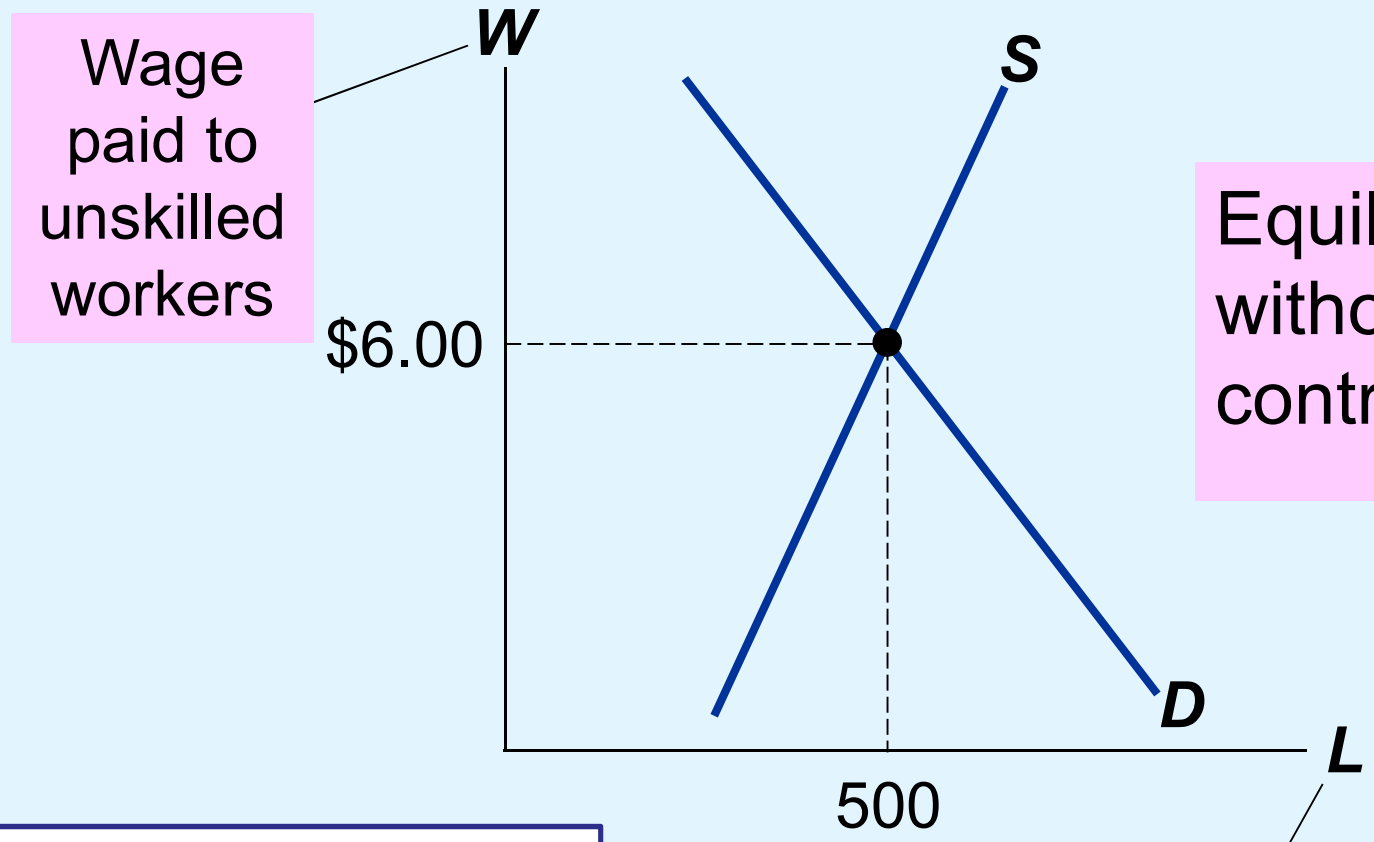
- Landlords: are not building new apartments and are failing to maintain existing ones
- People: more people want to find their own apartments and more people to move into a city
- Inefficient rationing mechanisms: long waiting lists; preference to tenants without children; discriminate on the basis of race; bribes to building superintendents



Shortages and Rationing

- Because of shortage
  - Sellers must ration the goods among buyers
- Some rationing mechanisms:
    - Long lines
    - Discrimination according to sellers' biases
  - Are often unfair and inefficient
    - The goods do not necessarily go to the buyers who value them most highly

EXAMPLE 2: The Market for Unskilled Labor



Wage paid to unskilled workers

Equilibrium without price controls

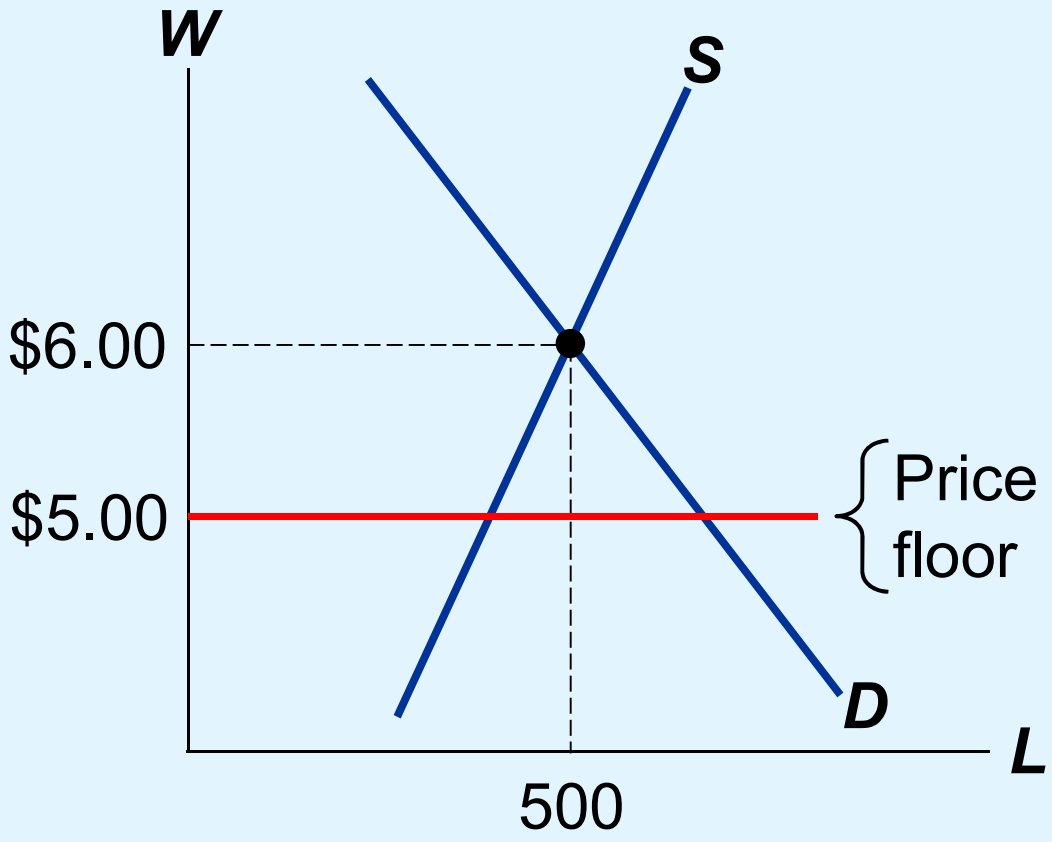
We focus on unskilled labor because the minimum wage is not relevant for higher skilled, higher wage workers.

Quantity of unskilled workers

The "price" of labor is simply the wage

How Price Floors Affect Market Outcomes

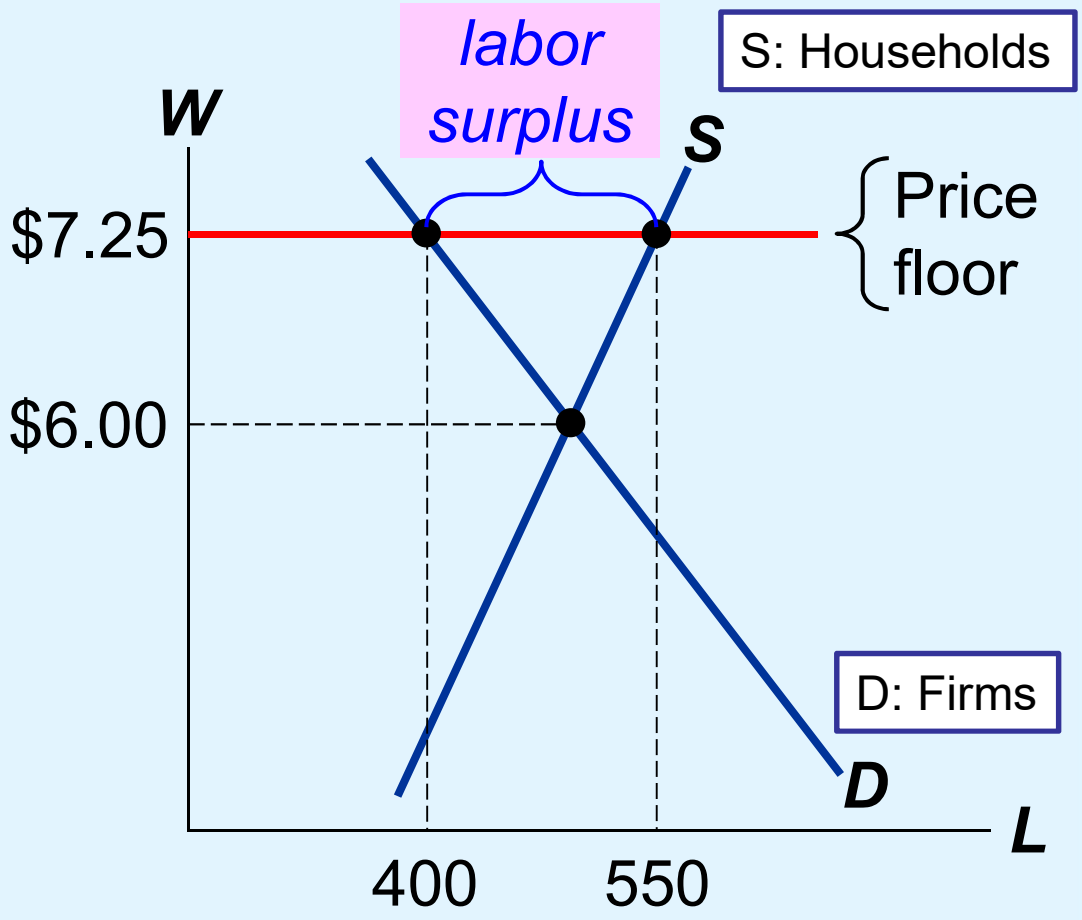
A price floor below the equilibrium price is **not binding** – has no effect on the market outcome.



How Price Floors Affect Market Outcomes

The equilibrium wage ($6) is below the floor and therefore illegal.
The price floor is **binding**, causes a surplus (i.e., unemployment).

Minimum wage laws do not affect highly skilled workers. They do affect teen workers. A 10% increase in the minimum wage raises teen unemployment by 1–3%.



Question

Lovers of classical music persuade Congress to impose a price celling of $40 per concert ticket.

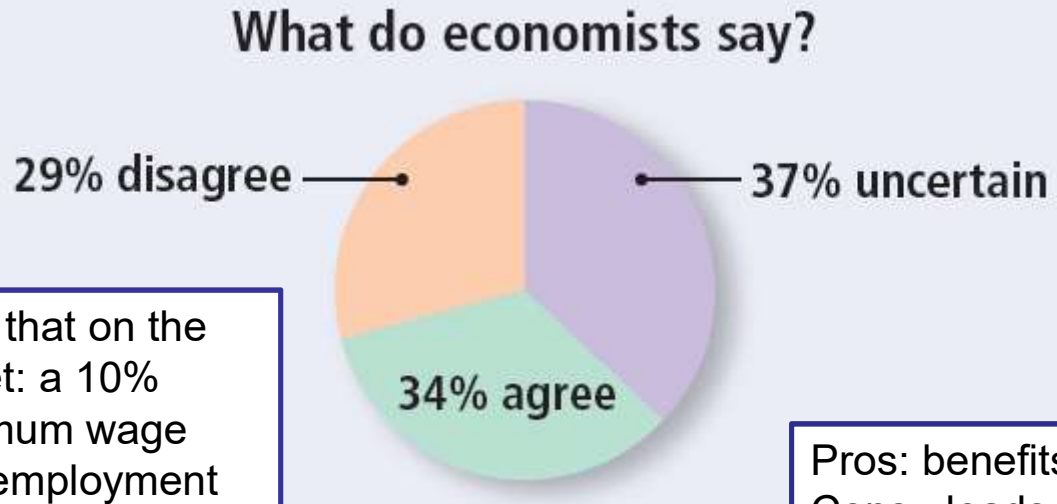
As a result of this policy, do more or fewer people attend classical music concerts? Explain.



ASK THE EXPERTS

The Minimum Wage

*"If the federal minimum wage is raised gradually to $15-per-hour by 2020, the employment rate for low-wage U.S. workers will be **substantially** lower than it would be under the status quo."*



Studies have shown that on the teenage labor market: a 10% increase in the minimum wage depresses teenage employment between 1 and 3%.

Pros: benefits the working poor
Cons : leads to unemployment

Active Learning 1

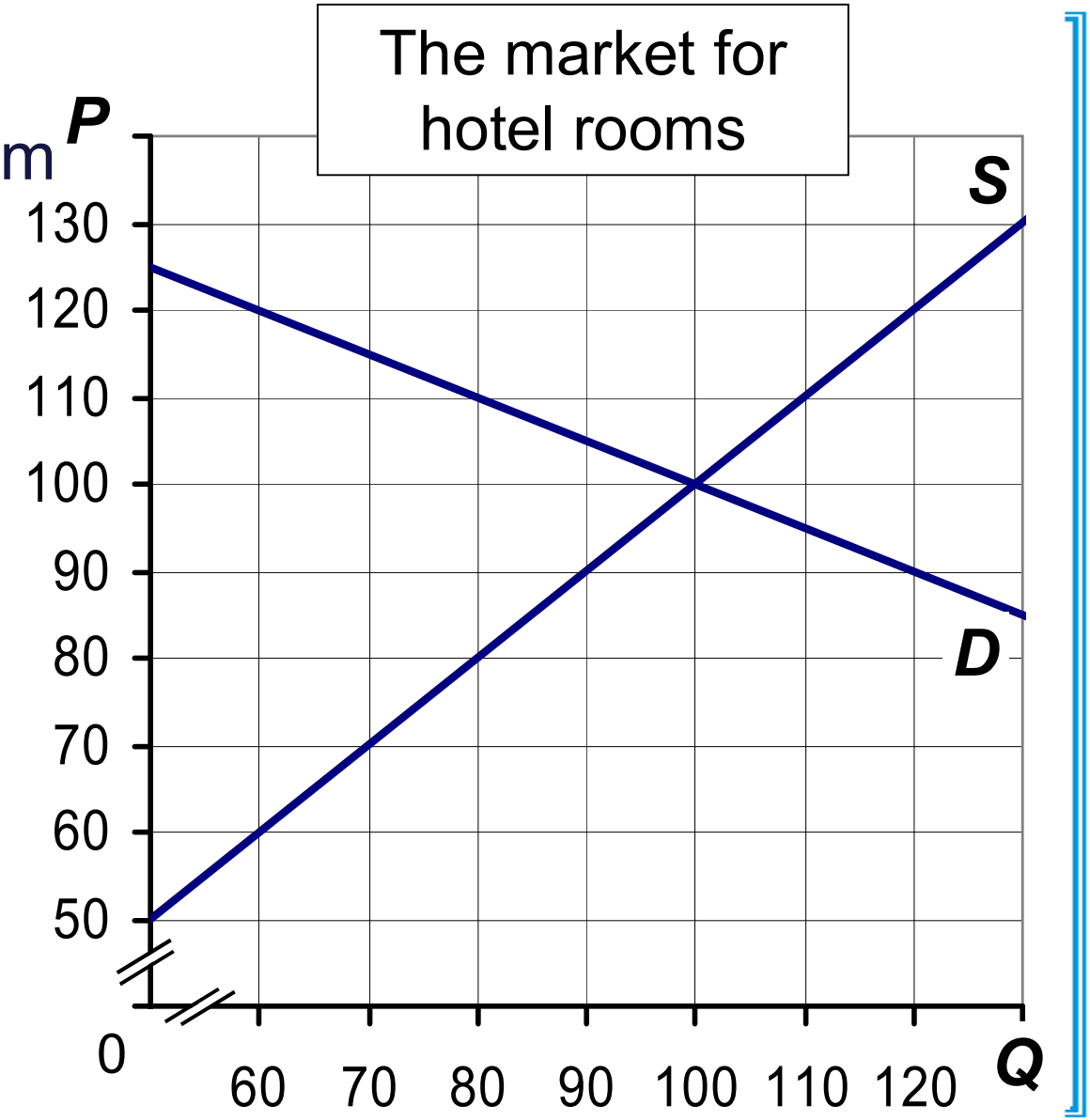
The market for hotel rooms is in equilibrium as in the graph.

- Determine the effects of:

A. $90 price ceiling

B. $90 price floor

C. $120 price floor

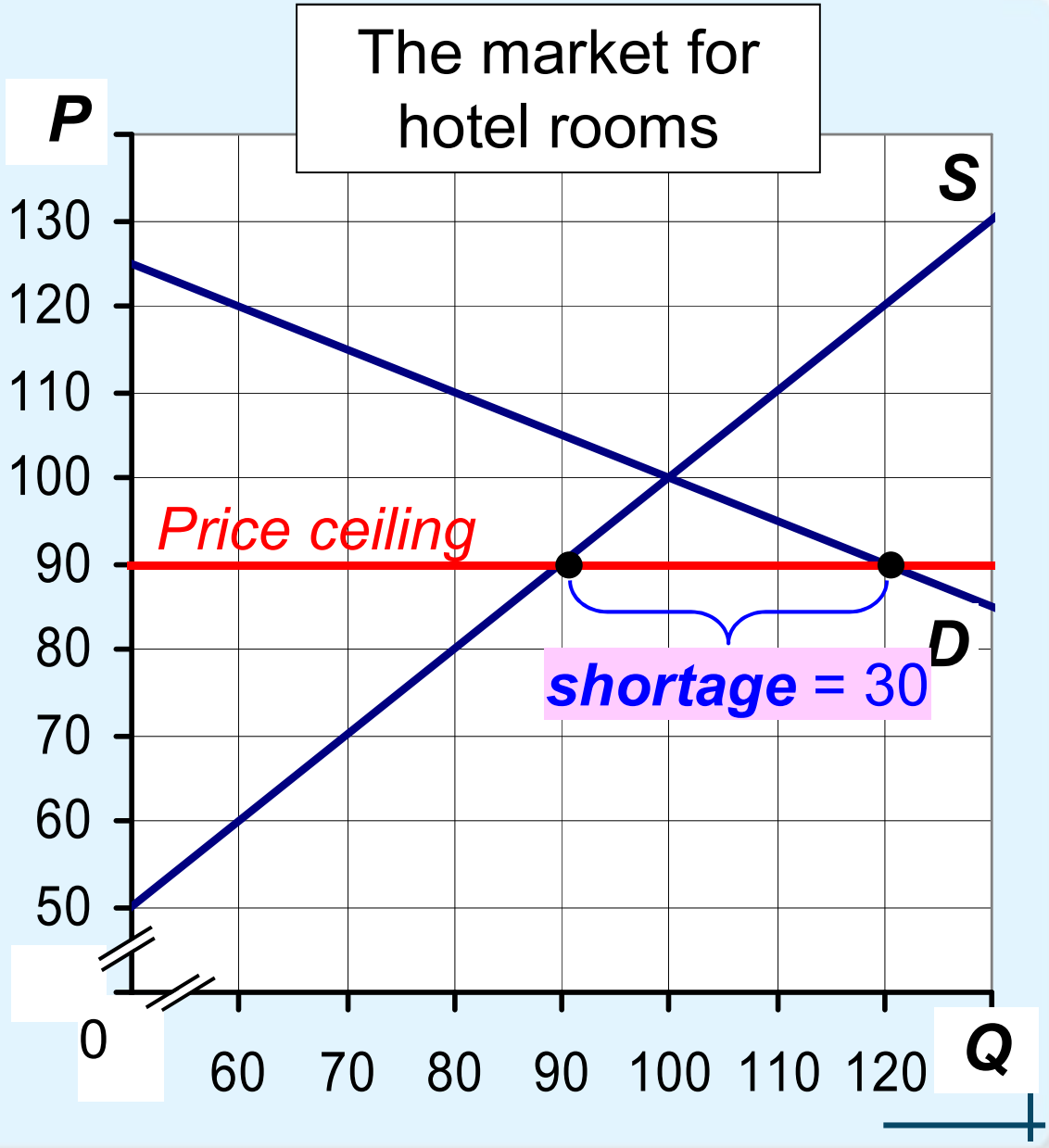


Active Learning 1

A. $90 price ceiling

The price falls to $90. (binding price ceiling below the equilibrium)

Buyers demand 120 rooms, sellers supply 90, leaving a shortage.

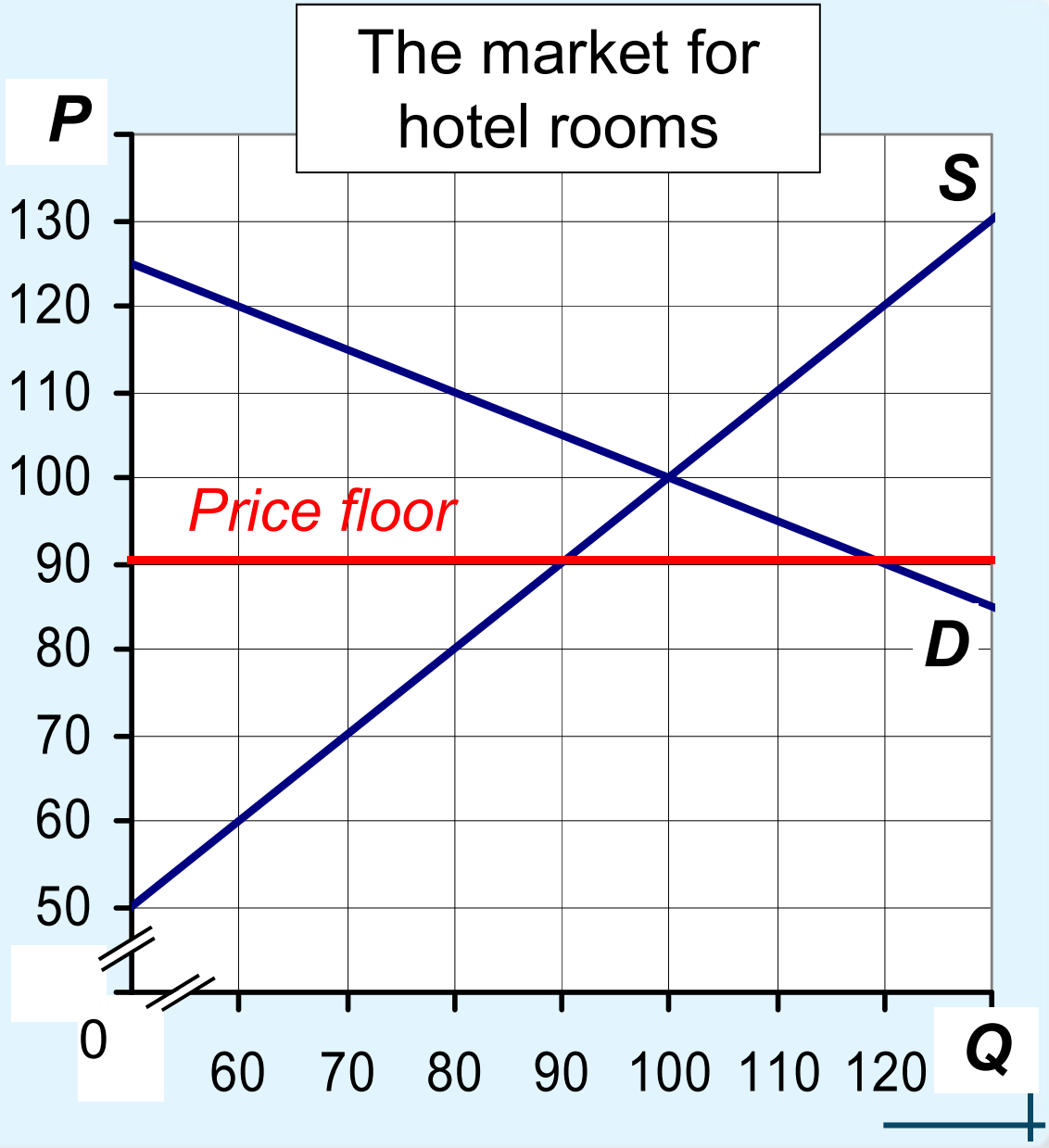


Active Learning 1

B. $90 price floor

Equilibrium price is above the $90 price floor, so the price floor is not binding.

P = $100,
Q = 100 rooms.

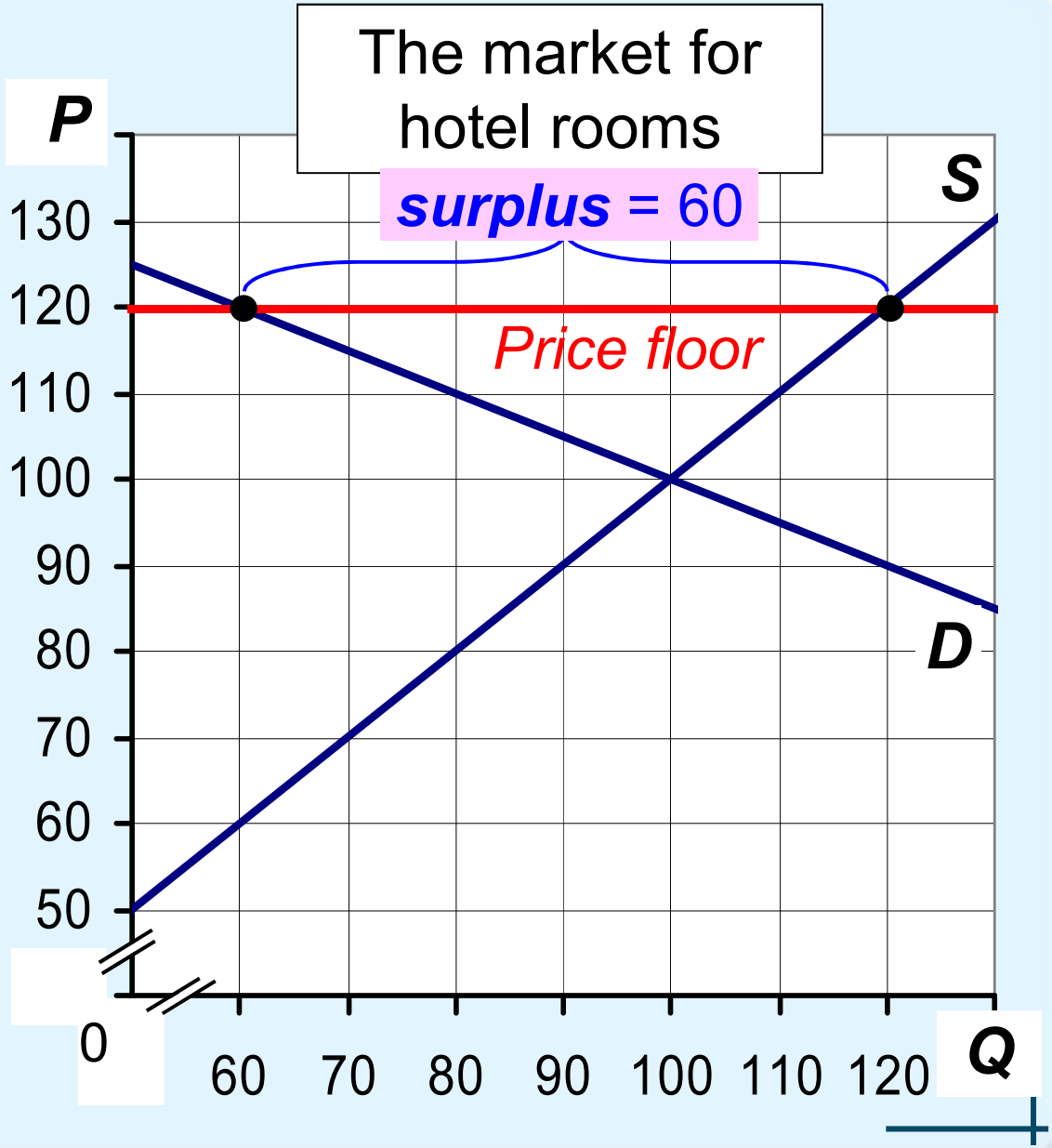


Active Learning 1

C. $120 price floor

The price rises to $120. (binding price floor above the equilibrium)

Buyers demand 60 rooms, sellers supply 120, causing a surplus.

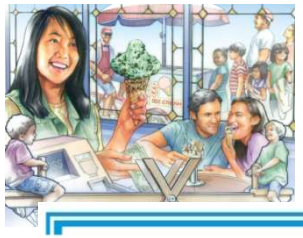


Question

Price per Frisbee	Quantity Demanded	Quantity Supplied
$11	1 million Frisbees	15 million Frisbees
10	2	12
9	4	9
8	6	6
7	8	3
6	10	1

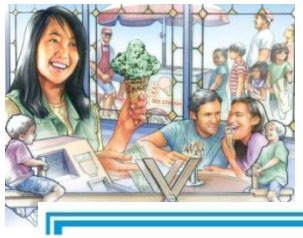


a. What are the equilibrium price and quantity of Frisbees?
b. A concerned Congress votes to impose a price floor $2 above the equilibrium price. What is the new market price? How many Frisbees are sold?
c. An concerned Congress votes to repeal the price floor and impose a price ceiling $1 below the former price floor. What is the new market price? How many Frisbees are sold?



Evaluating Price Controls

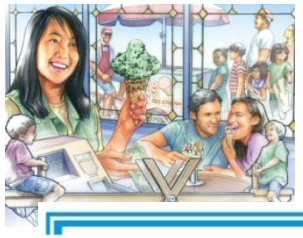
- Markets are usually a good way to organize economic activity
  - Economists usually oppose price ceilings and price floors
  - Prices are not the outcome of some haphazard process
  - Prices have the crucial job of balancing supply and demand
    - Coordinating economic activity



Evaluating Price Controls

- Governments can sometimes improve market outcomes
  - Want to use price controls
    - Because of unfair market outcome
    - Aimed at helping the poor
  - Often hurt those they are trying to help
  - Other ways of helping those in need
    - Rent subsidies
    - Wage subsidies (earned income tax credit)

EITC (勞動所得稅扣抵制): Workers receive a credit equal to a percentage of their earnings, up to a maximum amount. The credit increases with income up to a certain threshold, stabilizes, and then phases out.



Rent subsidies

租金補貼 全新進化

	300億租金補貼	舊租金補貼
所得限制	勝 每人每月平均所得低於最低生活費 3 倍可申請 (ex:台北市56046元)	每人每月平均所得低於最低生活費 2.5 倍可申請 (ex:台北市44170元)
申請限制	勝 申請人、配偶、未成年子女未擁有自住宅	申請人、配偶、戶籍內直系親屬、申請人配偶戶籍內直系親屬、申請人或其配偶孕有之胎兒、申請人父母均已死亡，且其戶籍內需要照顧之未成年或身心障礙兄弟姊妹未擁有自住宅
公益出租人	持續優惠 每屋每月租金收入免稅額 最高1萬5000元	

內政部

補貼戶數10萬戶

依照各地方分區分級補貼

第一級

一、家庭成員2人以上，且成員中具有低收入戶身分者

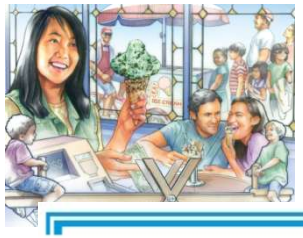
二、家庭成員3人以上，且成員中具有中低收入戶身分者

第二級 非屬第1級及第3級條件者

第三級 家庭成員1人且未滿40歲，且不具有經濟或社會弱勢身分者

直轄市、縣(市)分級	補貼金額區間
臺北市	3,000元–8,000元
新北市	2,000元–5,000元
新竹縣、新竹市	2,400元–5,000元
桃園市	2,400元–5,000元
臺中市	2,000元–5,000元
臺南市	2,000元–4,000元
高雄市	2,000元–4,000元
其他縣市	2,000元–3,600元

內政部

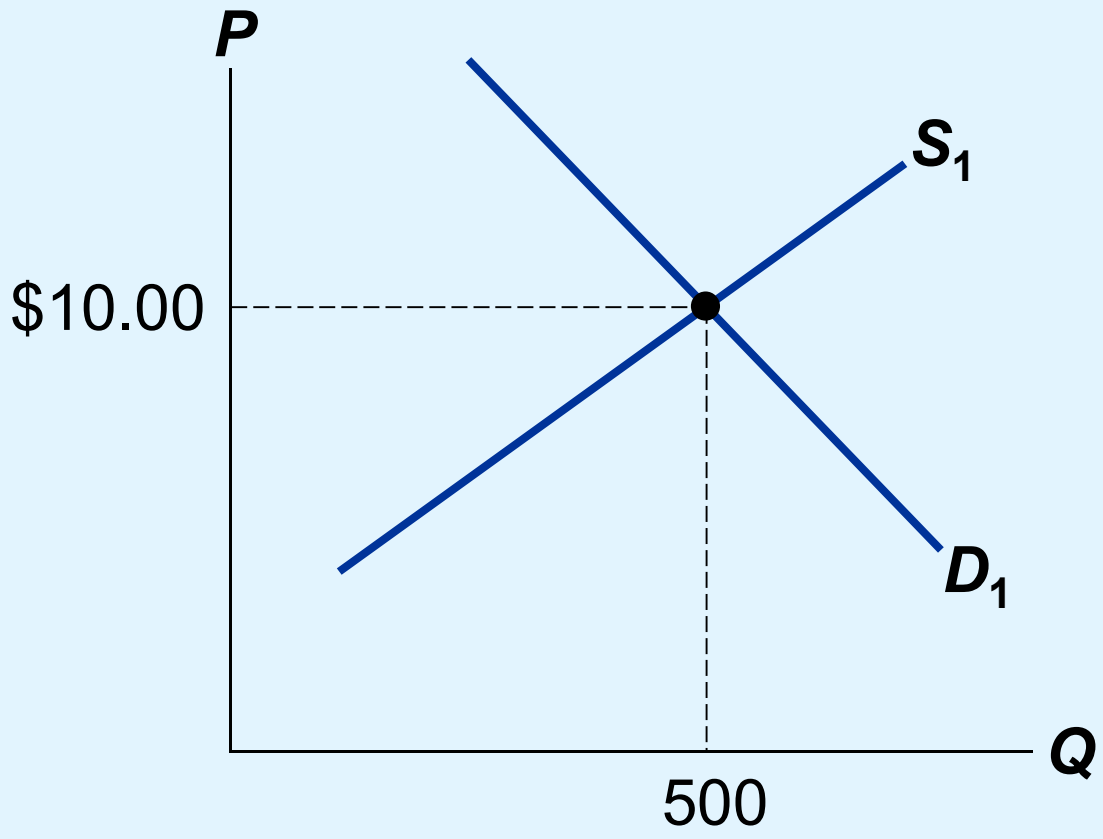


Taxes

- Government uses taxes
  - To raise revenue for public projects
    - Roads, schools, and national defense
- Tax incidence 租稅歸宿
  - Manner in which the burden of a tax is shared among participants in a market
    - The government can make the seller or the buyer to pay the tax

The tax can be a percentage of the good's price, or a specific amount for each unit sold. For simplicity, we analyze per-unit taxes only.

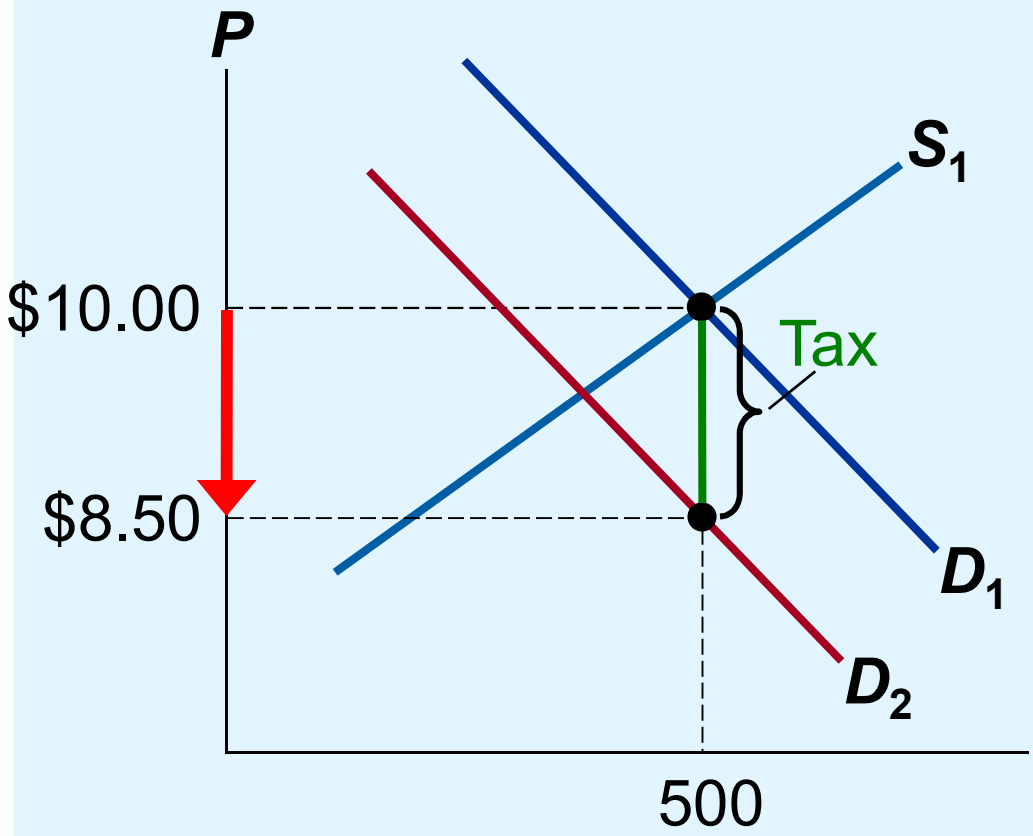
EXAMPLE 3: The Market for Pizza



Equilibrium without tax

A Tax on Buyers

Effects of a $1.50 per unit tax on buyers



Hence, a tax on buyers shifts the *D* curve down by the amount of the tax.

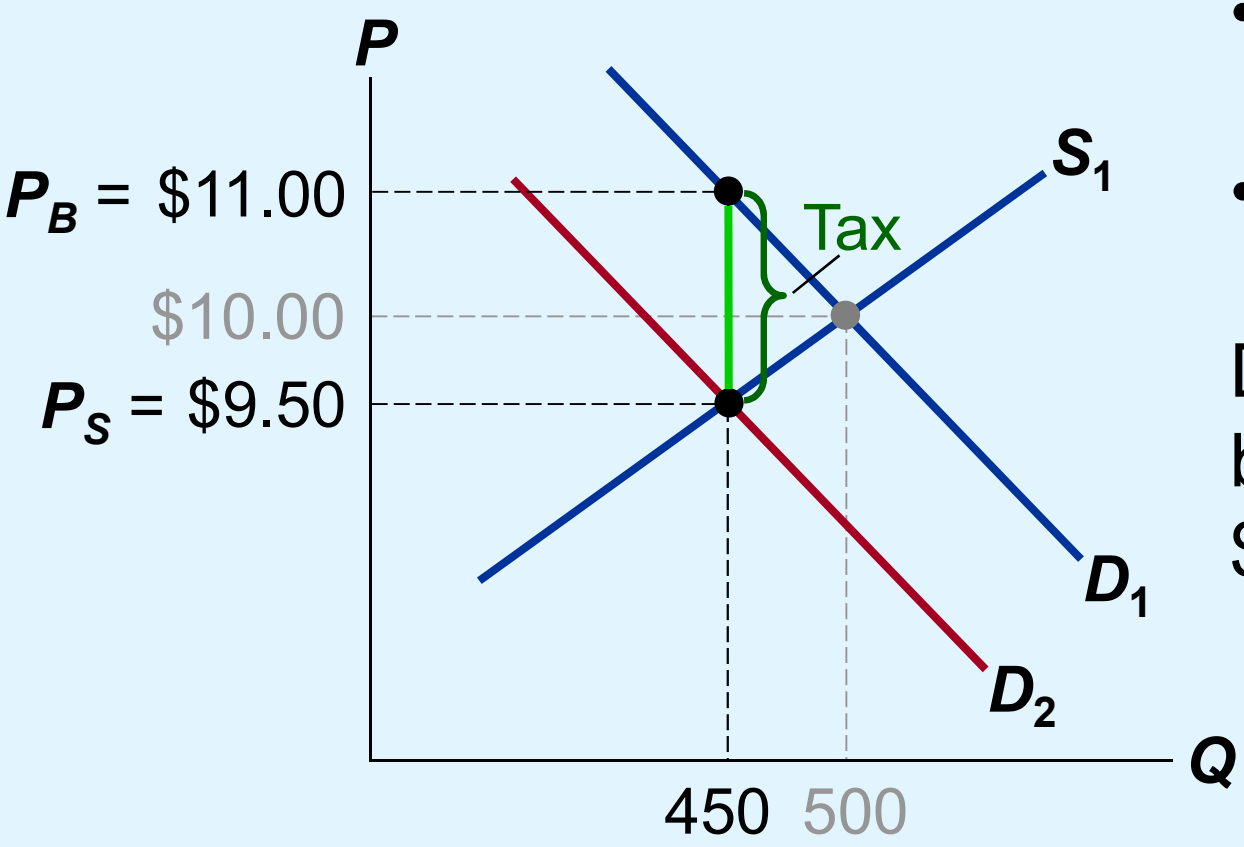
The price buyers pay is now $1.50 higher than the market price ***P***.

P would have to fall by $1.50 to make buyers willing to buy same ***Q*** as before.

- *E.g.*, if ***P*** falls from $10.00 to $8.50, buyers are still willing to purchase 500 pizzas.

A Tax on Buyers

Effects of a $1.50 per unit tax on buyers



New equilibrium:

- Q = 450
- Sellers receive P_S = $9.50
- Buyers pay P_B = $11.00

Difference between them = $1.50 = tax

How taxes on buyers affect market outcomes:

- Buyers and sellers share the burden of tax
- Sellers get a lower price, are worse off
- Buyers pay a lower market price, are worse off: Effective price (with tax) rises

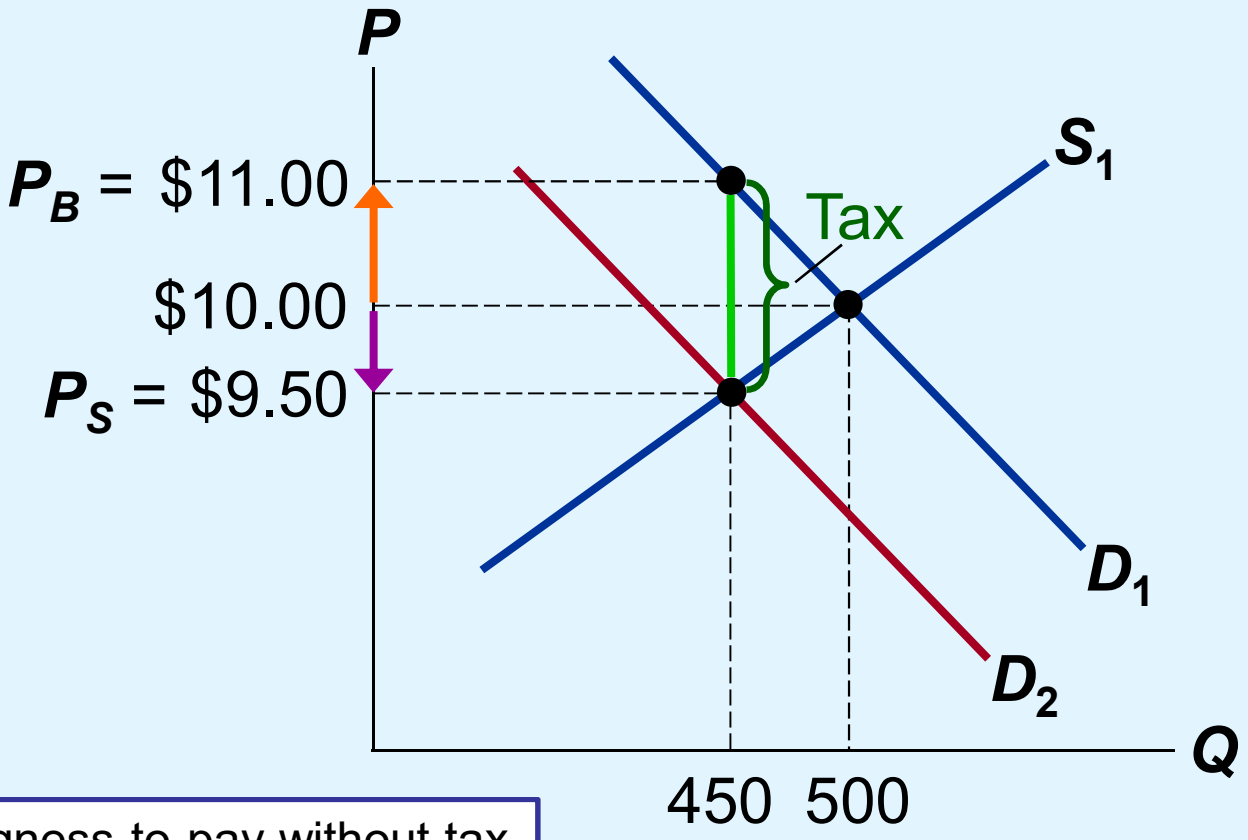
The **Incidence** of a Tax:

how the burden of a tax is shared among market participants

In our example,

buyers pay $1.00 more,

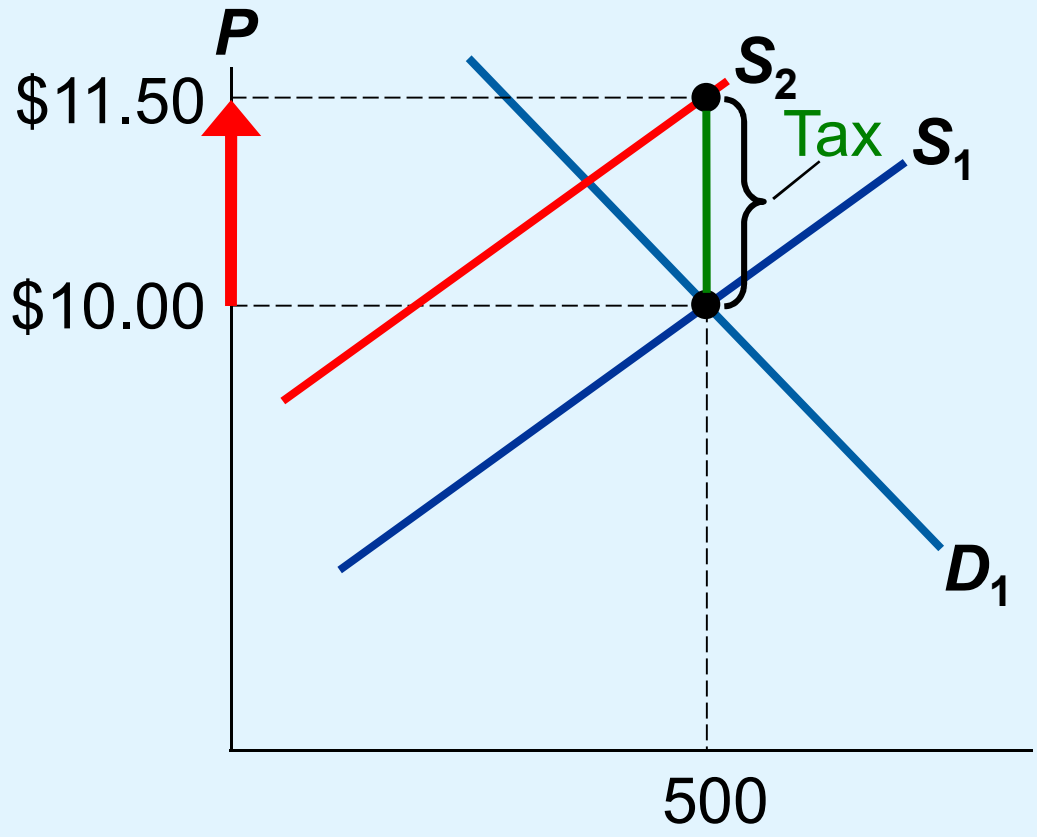
sellers get $0.50 less.



D_1: Consumers' willingness-to-pay without tax
D_2: Consumers' willingness-to-pay with tax

A Tax on Sellers

Effects of a $1.50 per unit tax on sellers



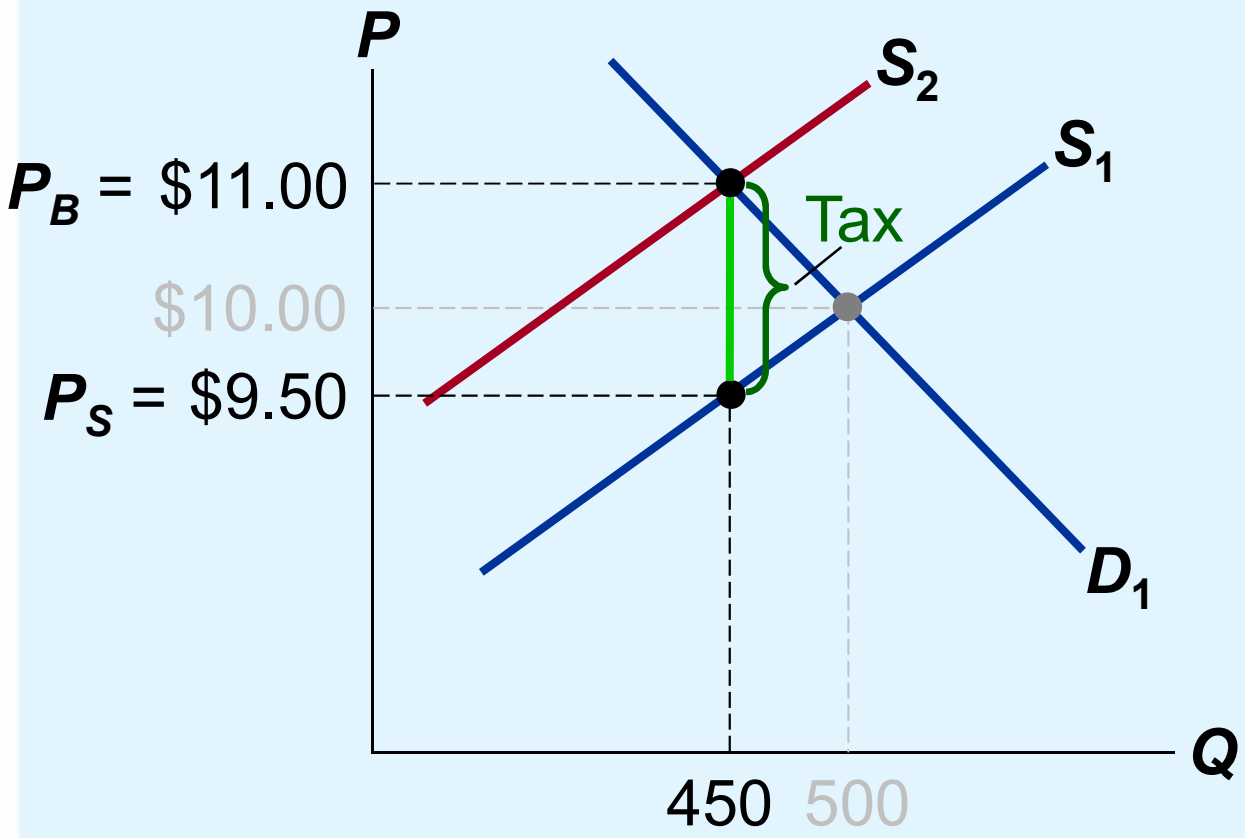
The tax effectively raises sellers' costs by $1.50 per pizza.

Sellers will supply 500 pizzas only if ***P*** rises to $11.50, to compensate for this cost increase.

Hence, a tax on sellers shifts the *S* curve up by the amount of the tax.

A Tax on Sellers

Effects of a $1.50 per unit tax on sellers



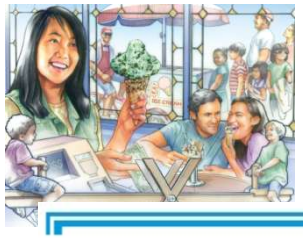
New equilibrium:

- Q = 450
- Buyers pay P_B = $11.00
- Sellers receive P_S = $9.50

Difference between them = $1.50 = tax

How taxes on sellers affect market outcomes:

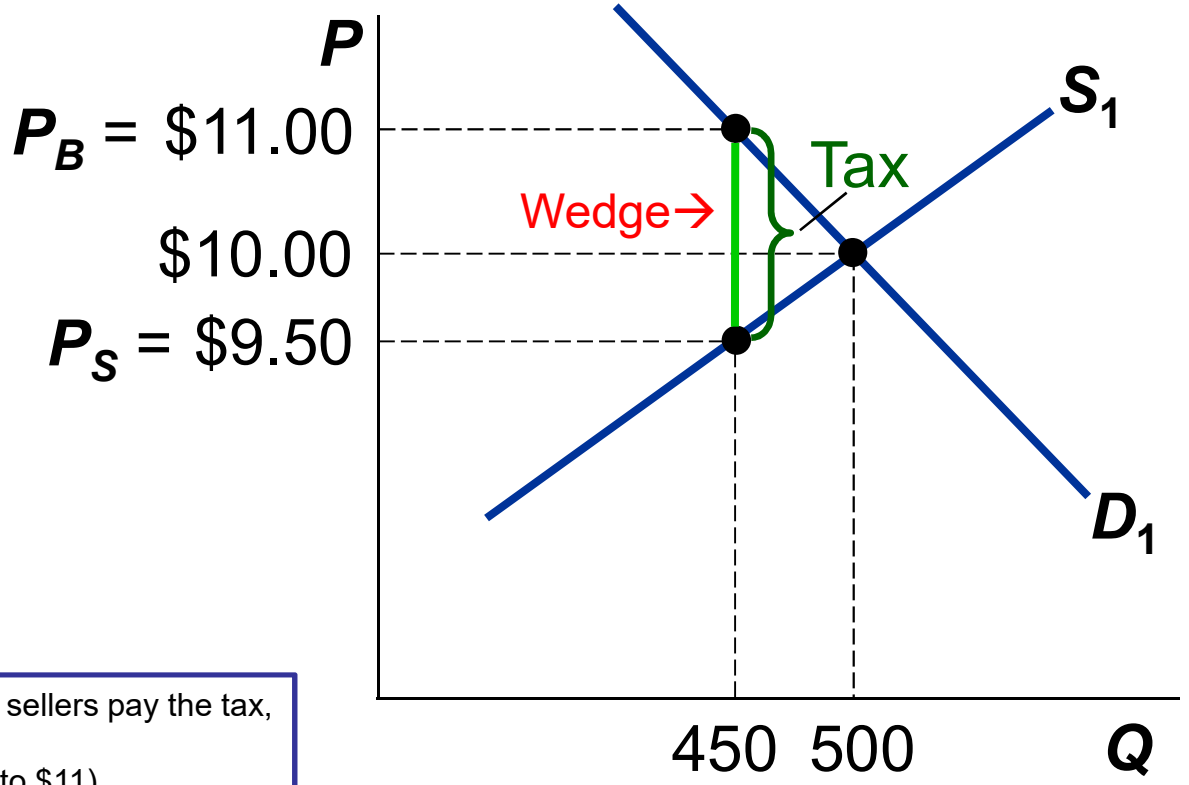
- Buyers and sellers share the burden of tax
- Sellers get a higher market price, are worse off: Effective price (deducted from tax) declines
- Buyers pay a higher price, are worse off



The Outcome Is the Same in Both Cases!

- The effects on P and Q, and the tax incidence are the same whether the tax is imposed on buyers or sellers!

A tax drives a wedge between the price buyers pay and the price sellers receive.



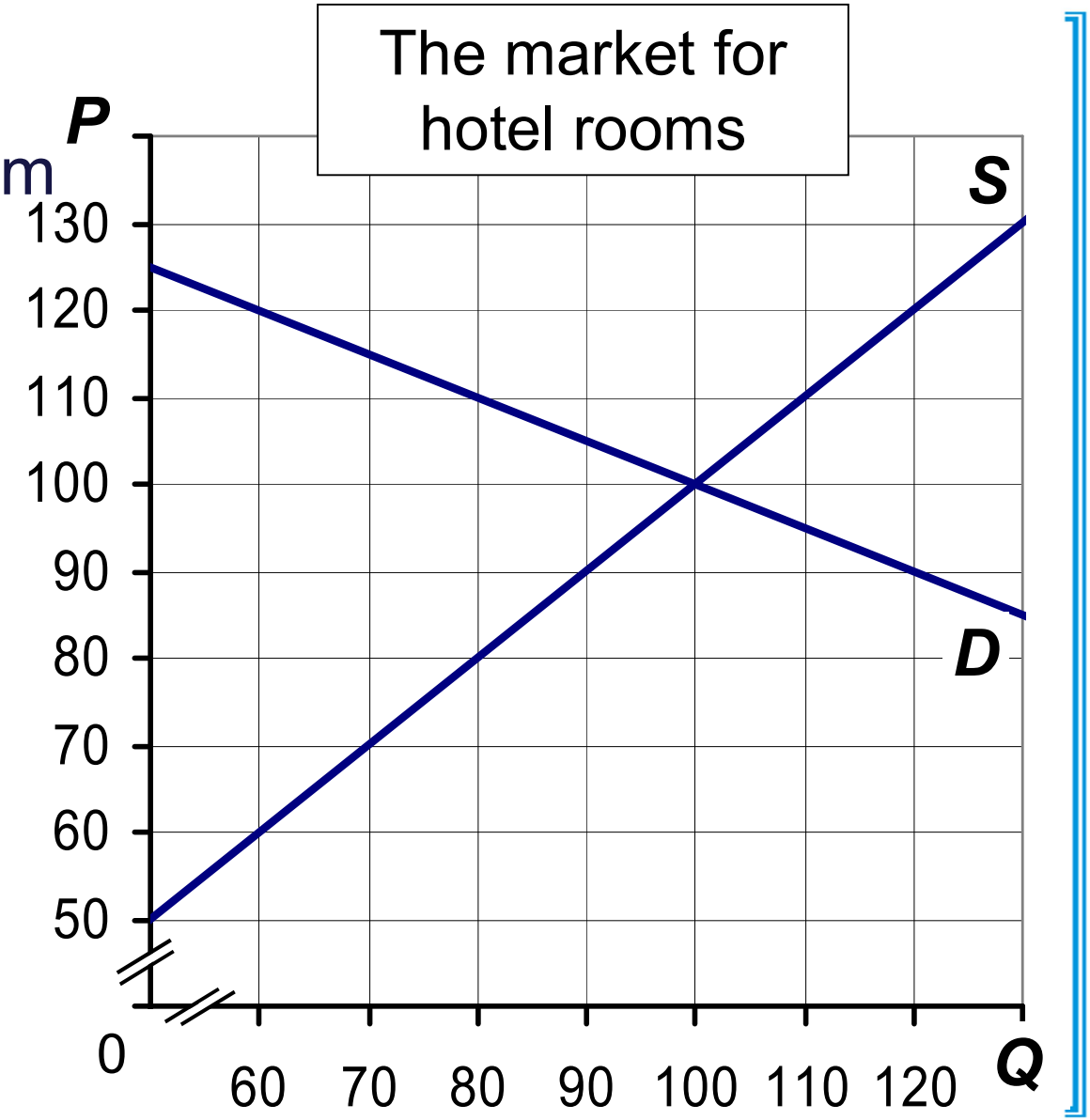
Whether the government makes buyers or sellers pay the tax, all of the effects are the same:
- the price buyers pay rises (in this case to $11)
- the price sellers receive falls (to $9.50)
- the equilibrium quantity falls (to 450)
- the incidence of the tax is the same (here, buyers pay $1 of the tax, while sellers pay $.50 of the tax on each unit)

Active Learning 2

Effects of a tax

The market for hotel rooms is in equilibrium as in the graph.

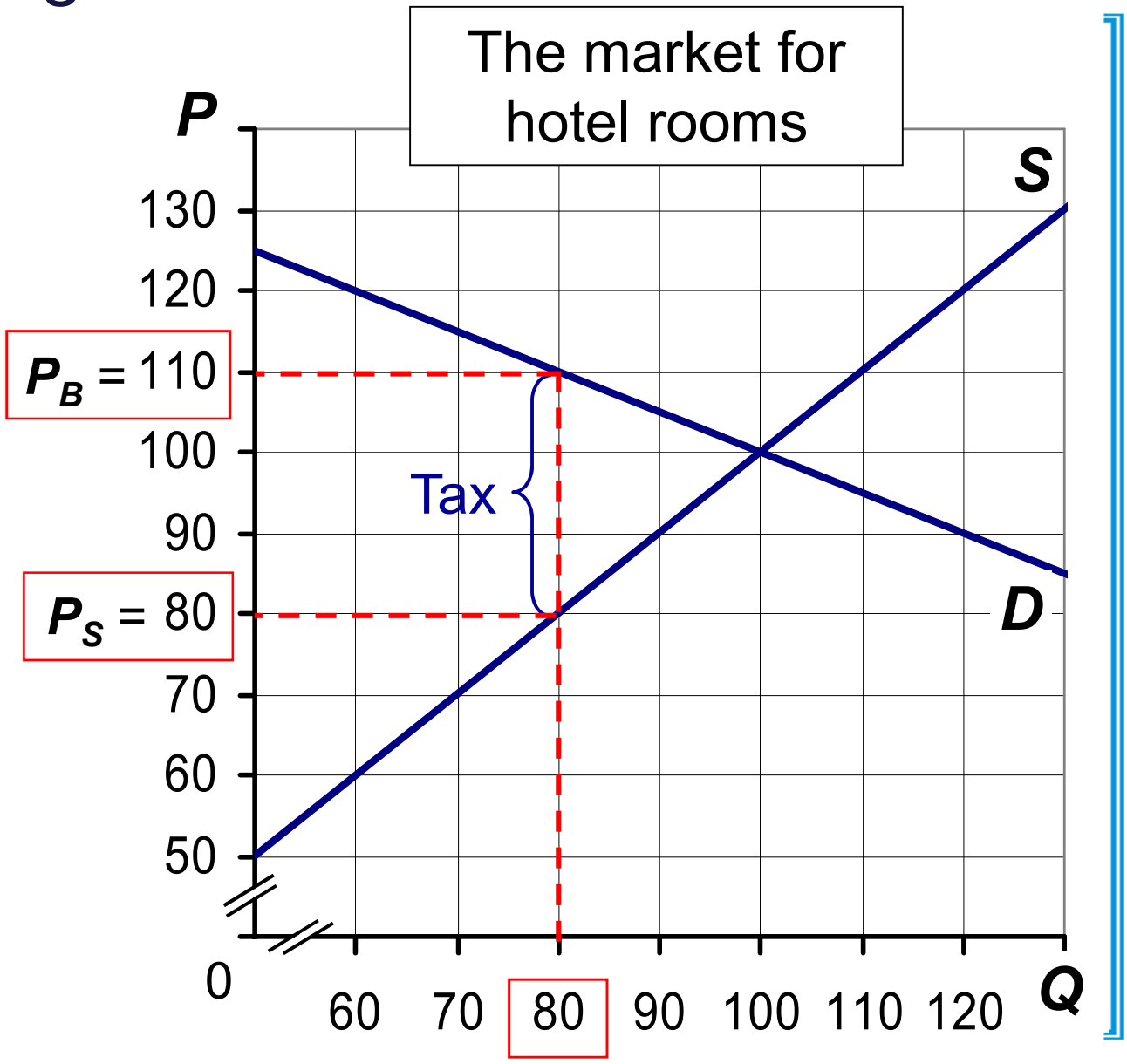
- Suppose the government imposes a tax on buyers of $30 per room
- Find the new Q, P_B, P_S, and incidence of tax.



Active Learning 2

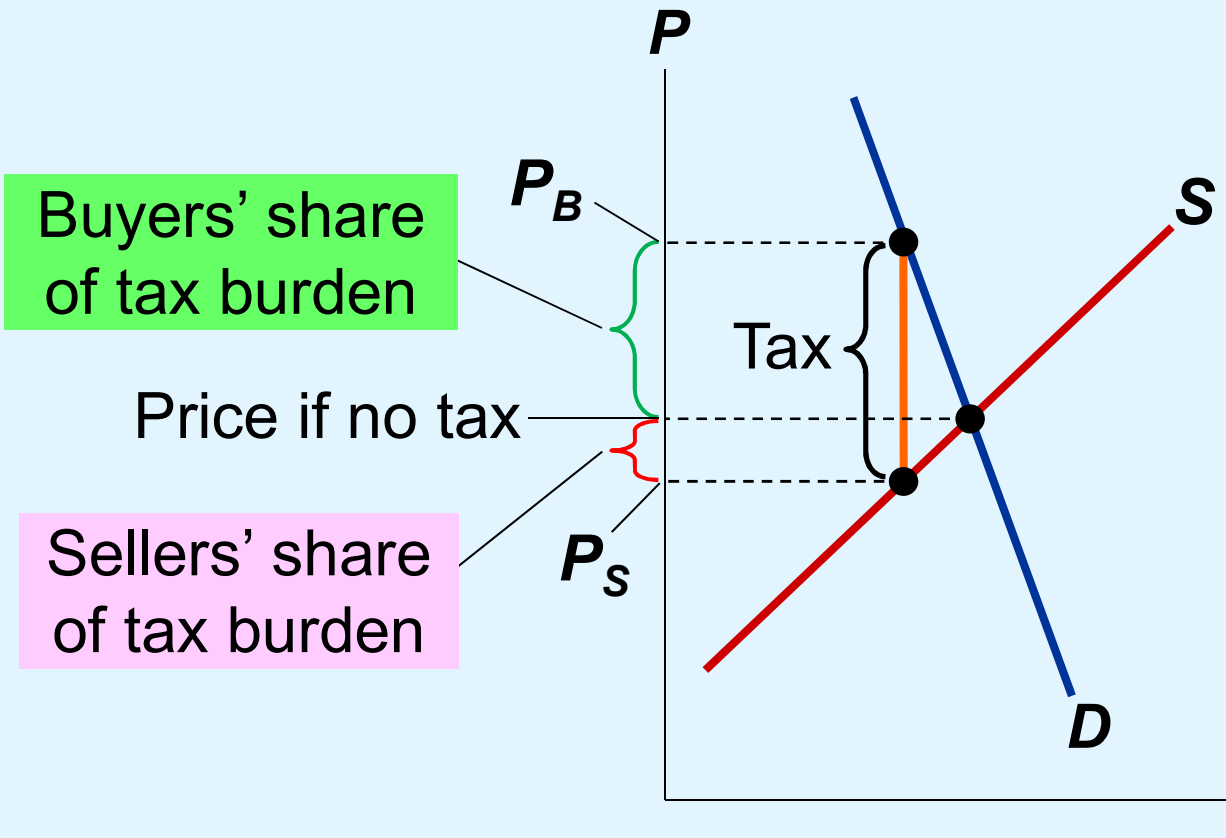
Answers

- Q = 80
- P_B = \$110
- P_S = \$80
- Incidence
  - buyers: \$10
  - sellers: \$20



Elasticity and Tax Incidence

CASE 1: Supply is more elastic than demand



Buyers' share of tax burden

Sellers' share of tax burden

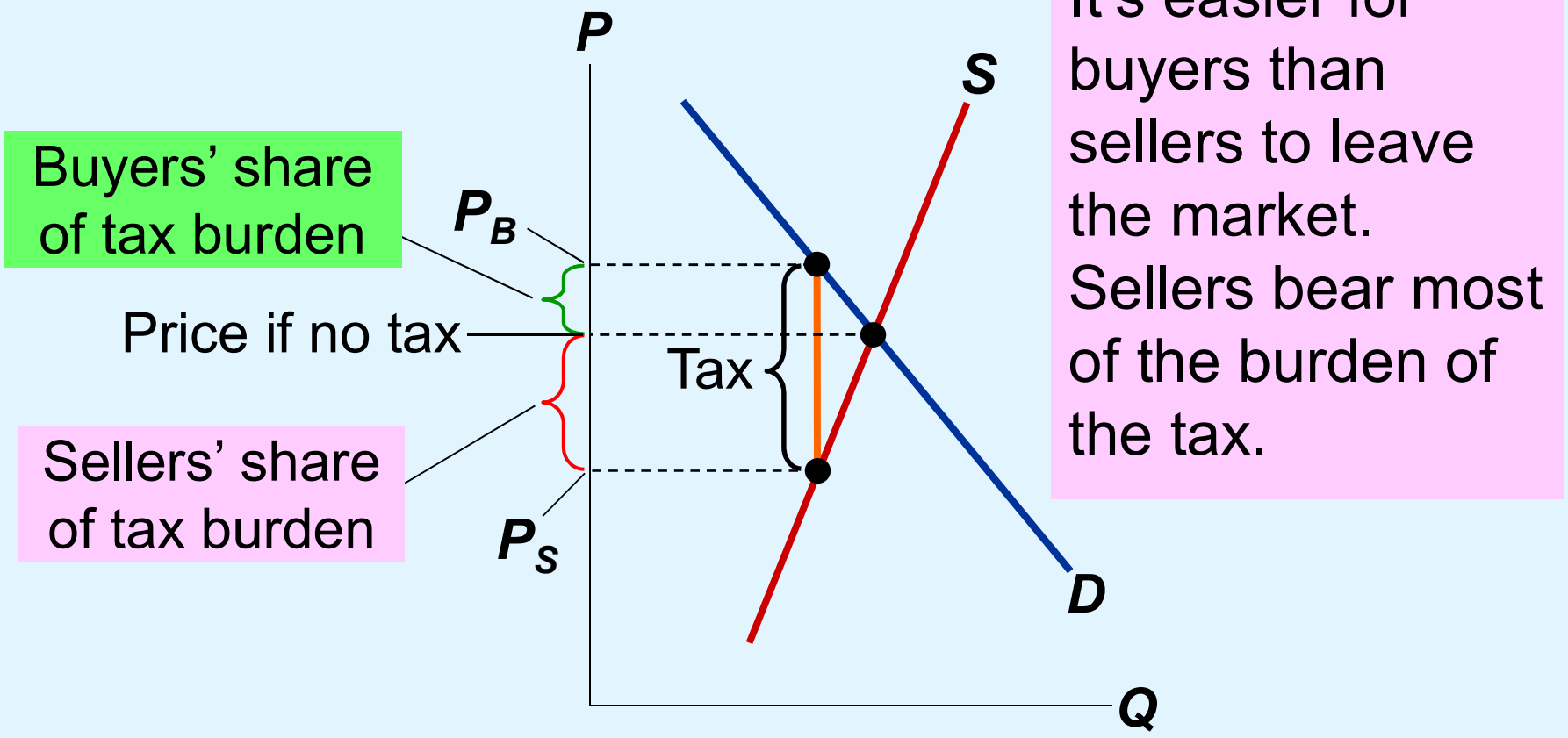
It's easier for sellers than buyers to leave the market.

So buyers bear most of the burden of the tax.

So what, then, does determine tax incidence? Turns out it's elasticity—specifically, the price elasticities of supply and demand.

Elasticity and Tax Incidence

CASE 2: Demand is more elastic than supply



It's easier for buyers than sellers to leave the market. Sellers bear most of the burden of the tax.

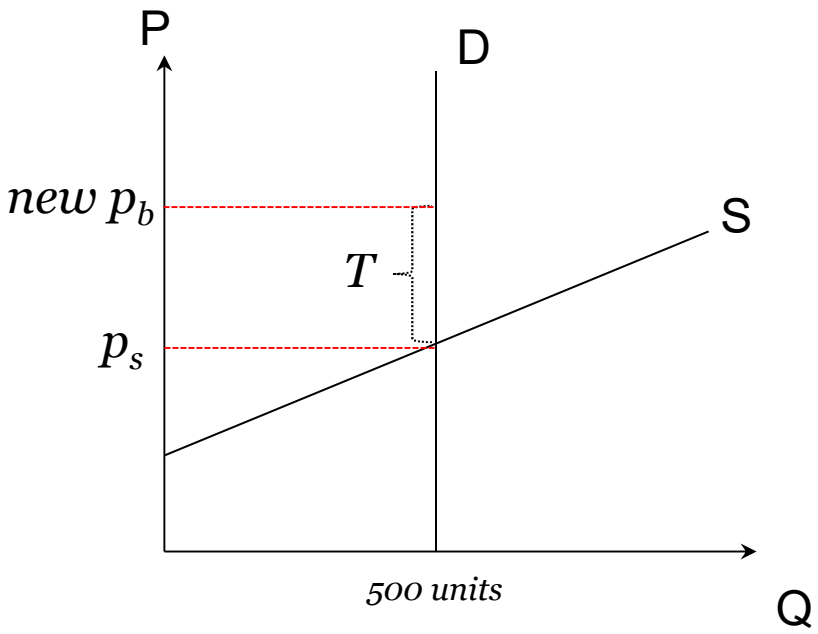
Tax burden falls more heavily on the side of the market that is less elastic:

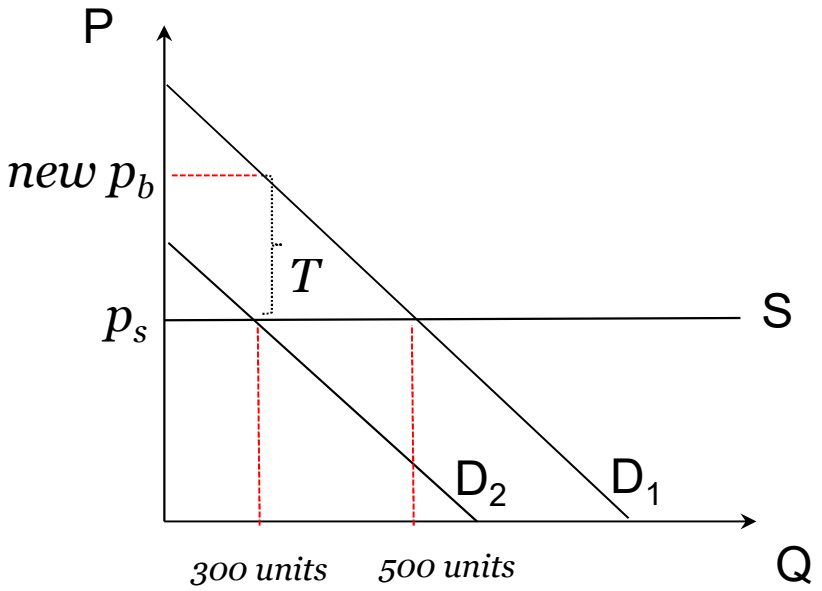
- Small elasticity of demand: Buyers do not have good alternatives to consuming this good
- Small elasticity of supply: Sellers do not have good alternatives to producing this good

Question

If the government places a $500 tax on luxury cars, will the price paid by consumers rise by more than $500, less than $500, or exactly $500? Explain.

Appendix Appendix





Let T = Tax
without Tax → p_b=p_s
with Tax → new p_b=p_s+Tax

Question

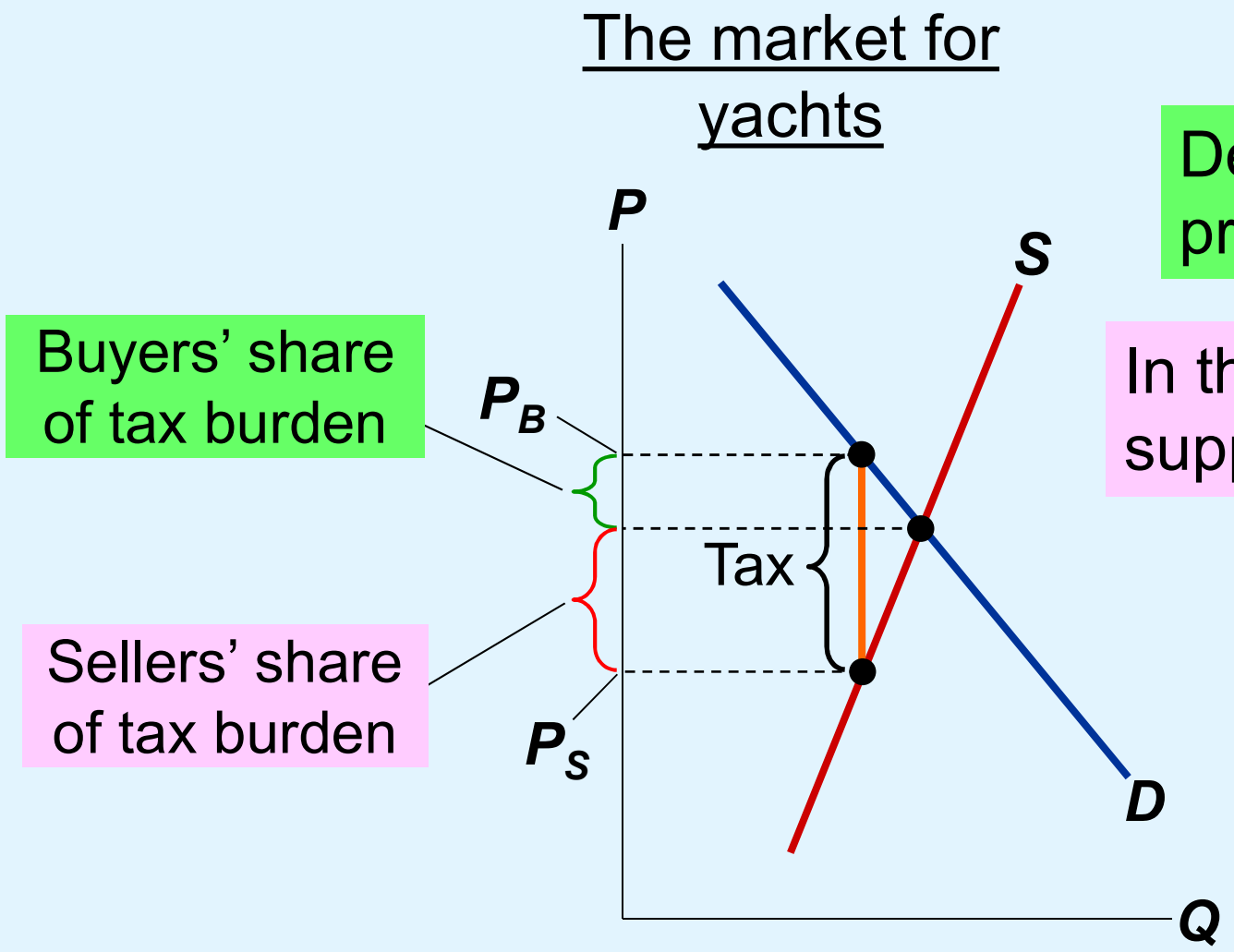
Congress and the president decide to impose a $0.50 tax on each gallon of gasoline sold.

a. Should they impose this tax on producers or consumers?
b. If the demand for gasoline were more elastic, would this tax be more effective or less effective in reducing the quantity of gasoline consumed?
c. Are consumers of gasoline helped or hurt by this tax?
d. Are workers in the oil industry helped or hurt by this tax?

Who pays the luxury tax?

- 1990, Congress adopted a new luxury tax
  - On yachts, private airplanes, furs, jewelry, expensive cars
  - Goal: to raise revenue from those who could most easily afford to pay
  - Luxury items
    - Demand is quite elastic
    - Supply is relatively inelastic

CASE STUDY: Who Pays the Luxury Tax?



Demand is price-elastic.

In the short run, supply is inelastic.

Hence, companies that build yachts pay most of the tax.

Most of the luxury tax in the US was repealed in 1993.

Active Learning 3 The 2011 payroll tax cut

Prior to 2011, the Social Security payroll tax (工資稅) was 6.2% taken from workers' pay and 6.2% paid by employers (total 12.4%). The Tax Relief Act (2010) reduced the worker's portion from 6.2% to 4.2% in 2011, but left the employer's portion at 6.2%.

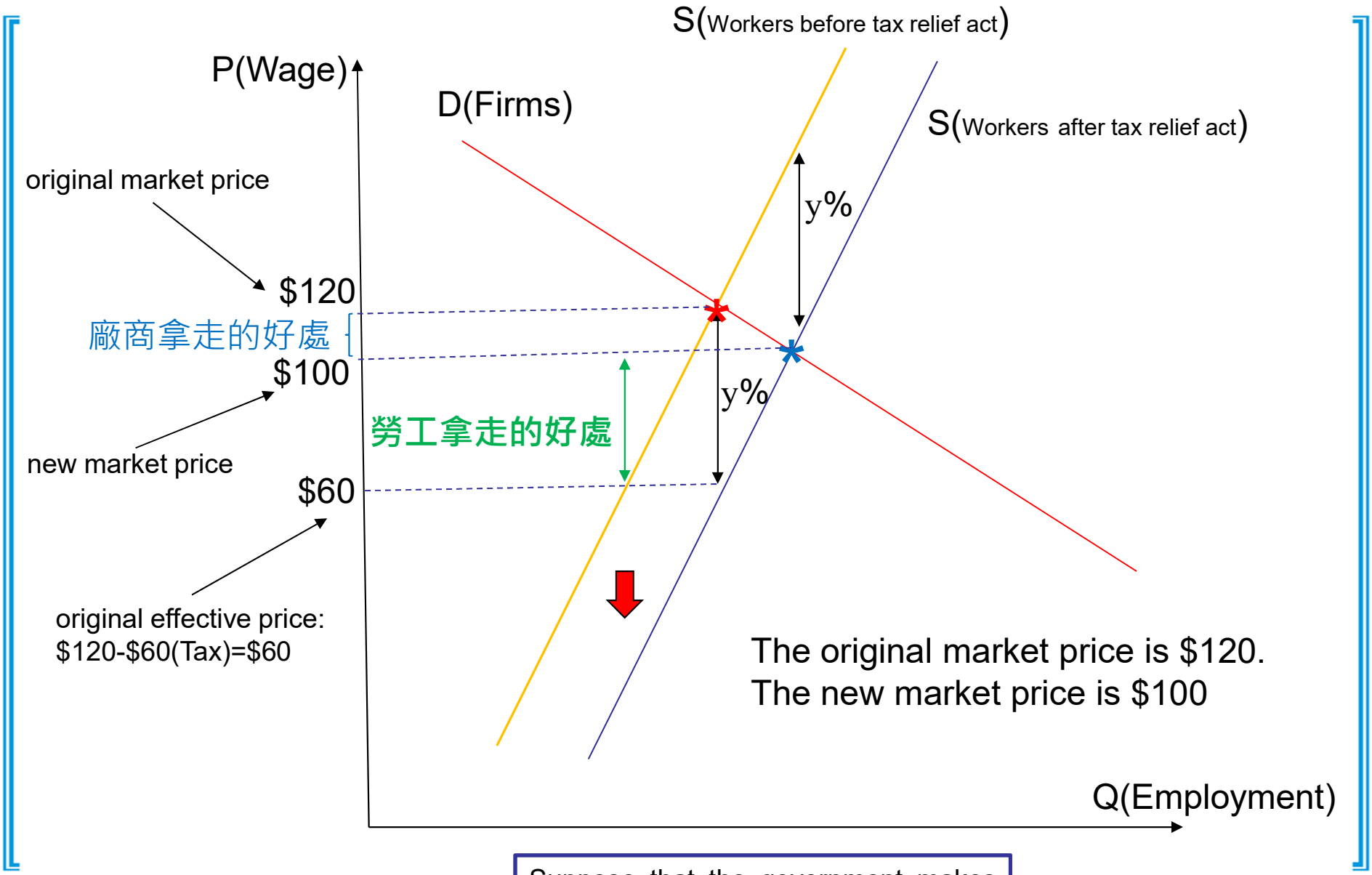
- Should this change have increased the typical worker's take-home pay by exactly 2%, more than 2%, or less than 2%? In general, do any elasticities affect your answer? Explain.
- FOLLOW-UP QUESTION: Who gets the bigger share of this tax cut, workers or employers? How do elasticities determine the answer?

Active Learning 3 Answers

- As long as labor supply and labor demand both have price elasticity > 0, the tax cut will be shared by workers and employers, i.e., workers' take-home pay will rise less than 2%.
- The answer does NOT depend on whether labor demand is more or less elastic than labor supply.

FOLLOW-UP QUESTION :

- If labor demand (資方) is more elastic than labor supply (勞方), workers get more of the tax cut than employers.
- If labor demand (資方) is less elastic than labor supply (勞方), employers get the larger share of the tax cut.



Suppose that the government makes the workers to pay the tax ($60).

Summary

- A price ceiling is a legal maximum on the price of a good. An example is rent control. If the price ceiling is below the equilibrium price, it is binding and causes a shortage.
- A price floor is a legal minimum on the price of a good. An example is the minimum wage. If the price floor is above the equilibrium price, it is binding and causes a surplus. The labor surplus caused by the minimum wage is unemployment.

In absence of market failures (which we will study in later chapters), the allocation of resources resulting from the free market equilibrium is optimal. Hence, government policies that alter this allocation tend to make the economy worse off.

Summary

- A tax on a good places a wedge between the price buyers pay and the price sellers receive, and causes the equilibrium quantity to fall, whether the tax is imposed on buyers or sellers.
- The incidence of a tax is the division of the burden of the tax between buyers and sellers, and does not depend on whether the tax is imposed on buyers or sellers.
- The incidence of the tax depends on the price elasticities of supply and demand.

Question

A market is described by the following supply and demand curves:

$Q^S = 2P$
$Q^D = 300 - P$

a. Solve for the equilibrium price and quantity
b. What are the price, quantity supplied, quantity demanded, and size of the shortage or surplus?
   - If the government imposes a price ceiling of $90
   - If the government imposes a price floor of $90
   - Instead of a price control, the government levies a tax on producers of $30. As a result, the new supply curve is: $Q^S = 2(P-30)$

Can Congress Repeal the Law of Market Forces?



Eye on Price Regulation (1 of 2)

Can Congress Repeal the Law of Market Forces?

When Congress enacts a new law and the President signs it, the outcome is not always what was intended.

A mismatch between intention and outcome is almost inevitable when a law or regulation seeks to block the law of market forces.

You've seen that the federal **minimum wage** law leaves some teenagers without jobs.

Problems would also arise if a law tried to cap **executive pay**.

Eye on Price Regulation (2 of 2)

Can Congress Repeal the Law of Market Forces?

Placing a cap on executive pay would work like putting a ceiling on home rents.

The quantity of executive services supplied would decrease and the most talented executives would seek jobs with the unregulated employers.

The firms in the most difficulty would face the added challenge of **recruiting and keeping competent executives and directors**.

The deadweight loss from this action would be large.